DEPARTMENT OF MATHEMATICS, IIT MADRAS

MA5950 — Mathematical Finance: Theory & Implementation

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Lecture 17: Martingales in Discrete Time

Information & Filtrations · The Fair Game · Why Backward Induction Works

September 8, 2026

Today's Agenda

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4. The Fundamental Theorem of Asset Pricing

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3. Discounted prices are martingales under Q
4. The Fundamental Theorem of Asset Pricing
5. Why backward induction works

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What Do We Know, and When?

At $t=0$ the album holds one photo: today's price. At $t=1$, two. At $t=2$, three. It never loses photos – information only accumulates – but always holds strictly less than the full future. That growing album is a **filtration** $\{\mathcal{F}_n\}$.

Filtration and Conditional Expectation

Filtration

$\mathcal{F}_0 \subseteq \mathcal{F}_1 \subseteq \cdots \subseteq \mathcal{F}_N$: everything observable up to time n. In the tree, $\mathcal{F}_n$ tells you which node you're at.

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Remark

$\mathbb{E}^Q[X \mid \mathcal{F}_n]$ is a *random variable* – it depends which node you're at. That's why backward induction computes a whole *layer* at each step.

Definition

Discrete-Time Martingale

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Given all you know today, the best forecast of tomorrow is today's value.

The Fair Game

Betting on a Fair Coin

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Hold Rs. 50, bet Rs. 1 on a fair coin. Expected wealth after the flip? Exactly Rs. 50 – gain and loss cancel. A **martingale**. Bias the coin your way: *submartingale* (drifts up). Against you: *supermartingale*. The martingale is the knife-edge *fair game*.

The Central Result

Key Takeaway

With $\tilde{S}_n = \dfrac{S_n}{(1+r)^n}$, under the risk-neutral measure Q:

$$\mathbb{E}^Q\left[\tilde{S}_{n+1} \mid \mathcal{F}_n\right] = \tilde{S}_n$$

The same holds for any derivative: $\tilde{V}_n = V_n/(1+r)^n$ is a Q-martingale.

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Not an extra assumption – it *is* q's defining property, restated.

Verifying Node by Node

$S_0=100$, $u=1.2$, $d=0.9$, $r=5\%$, so $q=0.5$

Node	S_n	$\mathbb{E}^Q[S_{n+1} \mid \mathcal{F}_n]$	$\tilde{S}_n$	$\mathbb{E}^Q[\tilde{S}_{n+1} \mid \mathcal{F}_n]$

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Holds exactly, at every node.

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Try $p = 0.7$ instead of $q = 0.5$: $\mathbb{E}^P[S_1] = 0.7(120) + 0.3(90) = 111$. The discounted price now drifts *upward*. Only q makes discounted prices driftless – hence "the martingale measure."

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The market is **complete** (every derivative replicable) $\iff$ that Q is *unique*.

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The market is **complete** (every derivative replicable) $\iff$ that Q is *unique*. The binomial model is complete – which is exactly why Lecture 14's replication always worked.

Two Languages, One Statement

The Same Fact, Twice

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Economic: “no riskless profit from nothing.” **Probabilistic**: “a measure exists making discounted prices driftless fair games.” Week 5 proved a small instance by hand. The theorem says this is completely general – not a binomial coincidence.

Backward Induction, Justified

Since $\tilde{V}_n$ is a Q-martingale, $\mathbb{E}^Q[\tilde{V}_{n+1} \mid \mathcal{F}_n] = \tilde{V}_n$ rearranges to:

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$$V_n = \frac{1}{1+r}\Big[qV_{n+1,\text{up}} + (1-q)\,V_{n+1,\text{down}}\Big]$$

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$$V_n = \frac{1}{1+r}\Big[qV_{n+1,\text{up}} + (1-q)\,V_{n+1,\text{down}}\Big]$$

Exactly Lecture 16's recursion. Backward induction *is* the martingale property, node by node.

And the One-Shot Version

Iterating from N down to 0 (tower property):

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Today's price = discounted risk-neutral expectation of the final payoff.

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But note

Backward induction computes the same thing step by step – and, as Lecture 18 shows, it's the *only* version that survives early-exercise rights.

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- Under Q, **discounted** prices are martingales – just q's defining property restated.
- **FTAP**: no arbitrage $\iff$ a martingale measure exists; complete $\iff$ unique.
- Backward induction **is** the martingale property, applied node by node.

Check Your Understanding

Discuss (2 minutes)

1. S_0=50, u=1.1, d=0.9, r=4%: find q, verify $\mathbb{E}^Q[\tilde{S}_1 \mid \mathcal{F}_0] = \tilde{S}_0$.

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2. Why is the *undiscounted* stock price not a Q-martingale?
3. Fair coin: is wealth a martingale? What if win probability is 0.55?

Looking Ahead

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What if you can exercise *any time*?

American Options – Lecture 18